PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
ARUSHA REGION

FORM THREE TERMINAL EXAMINATION

CODE:041 **BASIC MATHEMATICS**

TIME: 3: 00 HRS **26th May 2026**

INSTRUCTIONS

1. This paper consists of section A and B, with a total of (14) fourteen questions.
2. Answer all questions
3. Each question in section A carries (06) marks, while each question in section B carries ten (1) marks.
4. All necessary working and answers for each question must be shown clearly.
5. Mathematical tables and non-programmable calculators may be used.
6. Communication devices and any unauthorized materials are not allowed in the examination.

SECTION A: (60 Marks)

1. (a) A student was given 36,000/= shillings for personal use. He/she spent the money as follows: -
   (i) One third for buying exercise books.
   (ii) Half of the remaining money for buying calculator
   (iii) The amount left was used for different expenses.

   How much money used in (i), (ii) and (iii).

   (b) In a class of 80 students, 52 play football and the rest play netball. Find

   (i) The percentage of students playing football
   (ii) The percentage of students playing netball.

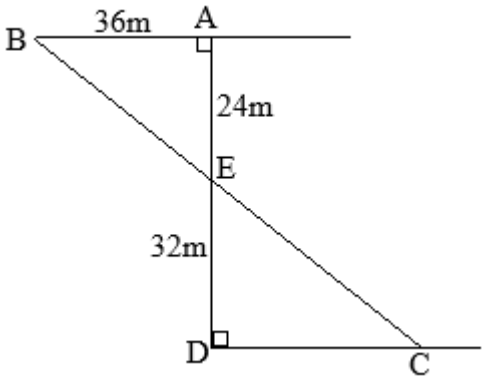
2. (a) Express $\frac{5+\sqrt{7}}{\sqrt{3}+\sqrt{7}}$ in the form of $a + b\sqrt{c}$, where a, b, and c are integers.
   (b) Given that $3log_3^x - log_3^{24} + 4 = 0$, find the value of x

3. (a) In a class of 100 students, 45 students study History, 40 students study Geography and five students study both subjects. Find the number of students who do not study of the two subject. (use the formula)
   (b) Given the universal set $\mu = \{15,30,45,60,75\}$ and B= $\{30,60\}$, find
   (i) (AUB)'
   (ii) Represent (AUB)' in Venn diagram

4. (a) Anna walked daily to school from home $2\,{}^{1}/_{2}$ kilometeres away. How many metres she will walk for 3 days.

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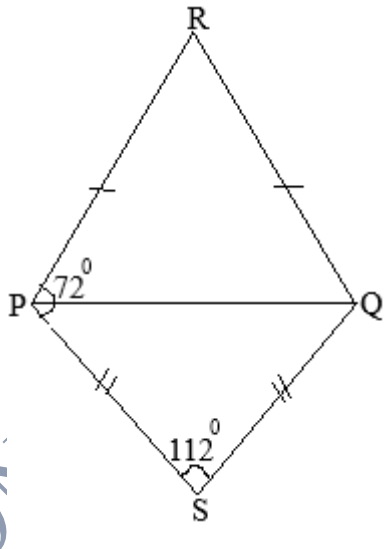
(b) You are given the figure below showing the school garden.



(i) Prove that the triangles AEB and CDE are similar.
(ii) Using similarity properties find $\overline{CD}$ in metres.

5. (a) The gradient of a straight line is -2, if it passes at points (2,-3) and (k,5). Find the value of k.
(b) In a rainy day 500 millilitres of water was collected in a tank after every minute. If it rained from 02:12pm to 05:04pm, find the amount of water collected in litres.

6. (a) In the figure below $S\hat{Q}P$. Find the size of R$\hat{Q}$S and $S\hat{P}Q = S\hat{Q}P$. Find the size of $R\hat{Q}S$



(b) A rectangular area is 72cm long and 40cm wide if a triangular area with base of 60cm has an area which is equal to the area of the rectangular area. Find the height of the triangular area.

7. (a) The T- shirt was bought at a discount of 20% during sales promotion day for 16,000 shillings. Find the original price of the T-shirt before discount
(b) The volume of a cylinder with a constant cross sectional area is directly proportional to its height. If the volume $V = 140cm^3$, when height, $h = 5cm$, find V when H = 1.5cm

(c) Express0.13^{0} in the form of $^{a}/_{b}$ where $b \neq 0$

8. (a) The sum of two consecutive even numbers are 46, find the numbers.

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(a) If 4 books and 8 pens costs 1240 shillings and 4 books and 3 pens costs 680 shillings. Find the price of a book and a pen.

9. (a) A rectangular plot of a land is 50m long and its diagonal is 57m. find the width of the plot. Leave your answer to two decimal places.

   (b) Without using mathematical tables or calculator find the value of $\frac{2Cos60^0}{3} + Sin\ 30^0 + \frac{Sin\ 45^0}{Cos\ 45^0}$

10. (a) By using general quadratic formula solve $2 - x - x^2 = 0$

    (b)Given that $T = \sqrt{\frac{(a-b)}{d}}$ make d the subject of the formula

SECTION B: (40 Marks)

11. The table bellow shows scores of 30 students in mathematics test.

Class interval	40 - 49	50 - 59	60 - 69	70 - 79	80 - 89	90 - 99
frequency	2	4	7	9	5	3

   (a) Calculate the median score correct to two decimal places.
   (b) Find the mean score
   (c) Draw a histogram and use it to estimate the mode.

12. The 20th term of an arithmetic progression is 60 and the 16th term is 20. Find;
   (a) The common difference.
   (b) The first term
   (c) The sum of the first 30 terms.

13. From a top of diff a boy sees two boats (A and B) on the sea. Boat A was X m, from the base of the diff and B was 10m far away from A. if the distance from the observing point to B was 50m and the angle of depression of the observer was 60º, find
   (a) The height of the diff
   (b) Distance of boat A to the base of the diff
   (c) Distance of boat B to the base of the diff.

14. Let the function f(x) be defined as $f(x) = \begin{cases} -4\ if\ x < -1 \\ x^2 + 1\ if\ -1 \leq x \leq 2 \\ 3\ if\ x^2 \geq 2 \end{cases}$

   (a) Sketch the graph of f(x)
   (b) State the domain and range of f(x)
   (c) Is f(x) one-to-one function? Why?

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